HAYA FATIMA

PETROLEUM AND PROCESS ENGINEER | DATA ANALYST

CONTACT

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Linkedin

GitHub

SKILLS

Technical Tools & Software

- PVTP Software
- MS Office
- JupyterLite Notebook
- Google Colab
- MATLAB
- Python
- SQL
- Power BI
- Ultimo
- TOMCAT
- P&IDs
- PFDs
- Data Analysis
- Data cleaning
- AI model exploration
- Data visualization

Soft Skills

- Teamwork
- Leadership
- Analytical Problem-Solving

LANGUAGES

- English (Fluent)
- Urdu (Native)
- Turkish (Basic)

PROFILE

Petroleum Engineering graduate (Gold Medalist, CGPA 3.906) with experience in operations, asset management, work order tracking, and data-driven reporting. Strong analytical and problem-solving skills with the ability to interpret operational data, identify discrepancies, and develop practical solutions. Proficient in Excel, Power BI, SQL, and Python, with experience in technical documentation, HSE compliance, and cross-functional coordination. Detail-oriented and adaptable, with a strong focus on accuracy, efficiency, and continuous learning.

WORK EXPERIENCE

Karachi Hydrocarbon Terminal, VTTI
NOV 2025- OCT 2026

Project Trainee| Operations Department

- Managed ULTIMO EAM work orders, updating records, tracking status, and supporting KPI reporting.
- Led asset tagging with Operations & Maintenance teams, improving asset identification and data accuracy.
- Maintained PTW records, tracking validity, closure, documentation, and HSE compliance.
- Prepared weight verification & G2G reports, analyzing data, identifying discrepancies, and supporting reporting.
- Prepared PFDs for the fire water system and terminal outside pipeline network.
- Developed a prototype AI Assistant chatbot for KHT Operations to provide quick access to operational knowledge.

Pakistan Petroleum Limited
05 AUG 2024 - 31 AUG 2024

Internee| Reservoir Modeling Department

- Worked in the Reservoir Modeling Department focusing on phase behavior analysis and datum pressure correction techniques.
- Assisted in interpreting fluid properties for reservoir simulations.
- Gained hands-on experience with reservoir engineering software and modeling workflows.
- Collaborated with engineers to understand the impact of pressure correction on reservoir performance forecasting.

Fuel Research Centre (PCSIR)
Mar 2024 - Apr 2024

- Assisted in laboratory fuel testing procedures and sample preparation.
- Learned and followed standard safety protocols during experimental analysis.
- Observed research on fuel efficiency and combustion properties under supervision.

EDUCATION

NED University of Engineering and Technology
Aug2026 - Present

Masters in Engineering Management
Chemical and Process Management

NED University of Engineering and Technology
Nov 2021 - June 2025

Bachelor of Engineering in Petroleum Engineering
CGPA: 3.906/4.00 | 1st Position | Gold Medallist

PROJECTS

Enhanced PVT Modeling of Gas Condensate Reservoirs Utilizing PVTP Software and EOS Models for Optimal Reservoir Management
Industry Collaboration: [United Energy Pakistan Limited]

- Simulated phase behavior and fluid properties using PVTP.
- Optimized EOS parameters to improve accuracy in predicting reservoir performance.
- Integrated real-world field data for enhanced model validation.

CERTIFICATION AND COURSES

Emergency Response and Fire Fighting Training| SGS -2026
Process Hazard Analysis - HAZOP, LOPA & SIL Training | SGS -2026
Google Advanced Data Analytics Professional Certificate - 2025
Google Project Management Professional Certificate - 2025
Google AI Essentials Specialization Certificate - 2025
Power Generation through Nuclear Reactor - 2022
Industry Exposure & Seminars - 2022
Attended technical seminars and field visits including Mari Petroleum's Sujawal Field and Nordex Pakistan Wind Energy Facility.

INTERESTS

- Data Analytics
- Well Testing and Pressure Transient Analysis
- Drilling Engineering
- Renewable Energy Technologies
- Reservoir Engineering and Fluid Modelling
- Technical Writing and Science Communication
- Solving Logic Puzzles and Brain Teasers

EXTRACURRICULAR ACTIVITIES

British Airways Data Science Job Simulation on Forage - February 2026

- Completed a simulation focussing on how data science is a critical component of British Airways' success.
- Scraped and analysed customer review data to uncover findings.
- Built a predictive model to understand factors that influence buying behaviour.

BCGx Data Science Job Simulation on Forage - February 2026

- Performed customer churn analysis to identify key drivers and develop a strategic investigation framework.
- Analyzed and visualized customer data using Python, Pandas, and NumPy to uncover actionable insights and trends.
- Developed and optimized a Random Forest model, achieving 50% recall in customer churn prediction.
- Prepared an executive summary presenting actionable recommendations to support data-driven decision-making.

AI Assistant – AI Chatbot Prototype -2026

- Developed a prototype AI-powered chatbot using a structured knowledge base for context-specific information retrieval. Built and tested the backend in Jupyter Notebook, managed code and knowledge files through GitHub, deployed the frontend, and connected it to a Render-hosted backend API. Designed the system to be adaptable to different operational documents and knowledge bases.
- Python • Jupyter Notebook • GitHub • FastAPI • Render • HTML/CSS/JavaScript

SENC Society (2021-2022)

- Contributed to various creative and photography initiatives as a volunteer.

Social Community Work (2023)

- Contributed as a volunteer to social community work initiatives within the school, fostering a sense of civic engagement and social impact.